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AIM

To write a C program to perform Priority process scheduling.

Algorithm

Step 1: Start

Step 2: Declare a structure process with the attributes pid, bt, wt, tat and prio.

Step 3: Declare variables i, j, n, avgw, avgt and p of size 10.

Step 4: Read the number of processes n.

Step 5: Read process id, burst time and priority for each process and store it in p.

Step 6: Sort the array p based on the priority of the processes.

from $i = 0$ to n and from $j = 0$ to $n - i - 1$,

if priority of $j >$ priority of $j + 1$,

swap $p[j + 1]$ and $p[j]$

Step 7: Calculate the waiting time and turnaround time for each process

If process is the first one, then its waiting time = 0.

Else

Waiting time = sum of burst time and waiting time of all the processes that come before it.

Turnaround time = sum of waiting time and burst time.

Step 8: Calculate the average waiting time and average turnaround time for all the processes.

Step 9: Display process id, burst time, waiting time and turnaround time with the average waiting time and average turnaround time.

Step 10: Stop

Result

The program was executed successfully and the output was verified.